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BUILDScan Rover AI

Autonomous AI-Assisted Robotic System for Indoor Structural Health Inspection and Crack Detection

ROBOTICS FINAL PROJECT REPORT

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Abstract

Reinforced concrete civil infrastructure undergoes progressive degradation due to environmental exposure, chemical leaching, and mechanical loading. Timely identification and geometric quantification of surface cracks are paramount to prevent moisture ingress, internal rebar oxidation, and catastrophic failure. Conventional manual surveys remain subjective, labor-intensive, hazardous, and devoid of standardized digital audit trails. This report presents an empirical evaluation of **BUILDScan Rover AI**, a distributed cyber-physical robotic platform designed for automated indoor structural defect screening.

The architecture decouples lightweight mobile capture onboard a Raspberry Pi 4 (equipped with a Logitech Brio 100 optical sensor and an ESP32 microcontroller safety watchdog) from high-throughput neural inference and dashboard monitoring on an AI workstation (Dell G15 with an NVIDIA GeForce RTX 3050 Laptop GPU). Real-time instance segmentation is performed by an optimized `YOLO26n-seg` neural network with 3,053,055 parameters (checkpoint size: 6.23 MB). Embedded validation records demonstrate a bounding-box precision of 84.08%, recall of 71.49%, and mAP@50 of 78.39%; segmentation mask precision, recall, and mAP@50 reach 77.46%, 65.46%, and 67.86%, respectively. Inference executes in 14.08 ± 2.05 ms (71.06 FPS) on the workstation GPU with an 80.6 MB VRAM footprint, compared to 77.80 ms (12.87 FPS) on CPU fallback.

Complementing local perception, cloud visual reasoning via Google Gemini 3.8 Flash provides on-demand severity triage and civil remediation recommendations governed by strict Pydantic JSON schemas. A 17-point automated verification suite evaluates model loading, image immutability, stale-frame suppression, standby diagnostics, and certified PDF generation. Physical calibration limitations, monocular depth ambiguity, and systemic failure recovery are rigorously analyzed, establishing a verified foundation for autonomous structural health inspection.

Keywords: Structural Health Monitoring; Mobile Robotics; Crack Segmentation; YOLO26n-seg; Deep Learning; ROS 2 Humble; Edge Computing; Multimodal AI; Gemini Flash; Pinhole Camera Calibration.

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Chapter 1

Introduction

1.1 Background and Motivation

Civil infrastructure assets—such as multi-story buildings, parking structures, highway overpasses, and transit tunnels—are composed predominantly of reinforced concrete. Over time, concrete surfaces inevitably develop cracks originating from plastic shrinkage, thermal expansion, environmental carbonation, cyclic stress concentrations, and seismic excitation [1]. Hairline surface fissures (< 0.30 mm) often appear benign but act as capillary conduits for moisture, atmospheric carbon dioxide, and deicing chlorides. Once moisture reaches the embedded steel reinforcement, rust oxidation induces expansive volumetric pressures that cause severe internal spalling, delamination, and sudden structural compromise [4].

In current engineering practice, structural condition surveys depend almost exclusively on human inspectors equipped with handheld crack comparator cards, tape measures, scaffolds, and manual logbooks [5]. This traditional methodology exhibits critical limitations:

- **Subjectivity and Inconsistency:** Crack identification and aperture measurements exhibit significant intra- and inter-inspector variability, particularly in dimly lit or confined indoor environments.
- **Safety Hazards:** Manual inspection requires personnel to navigate confined crawl spaces, industrial voids, and elevated scaffolding beneath deteriorated slabs.
- **Absence of Digital Continuity:** Field notes and uncalibrated phone photographs lack synchronized spatial coordinates, temporal continuity, and standardized digital defect registries.

1.2 Problem Statement

Executing deep learning-based image segmentation directly on compact mobile robot single-board computers (such as the Raspberry Pi 4) presents severe thermal and computational bottlenecks. Standard PyTorch neural inference on edge CPUs yields latencies exceeding 400 ms per frame (< 2.5 FPS), leading to frame backlog and buffer overflow. Conversely, transmitting raw, uncompressed 1080p video across local Wi-Fi networks saturates wireless bandwidth and introduces jitter.

An effective cyber-physical system must therefore decouple low-level robotic motion, sensor acquisition, and hardware safety from high-throughput neural inference, multimodal reasoning, and user telemetry.

1.3 Project Objectives

The BUILDScan Rover AI project addresses this challenge through five concrete objectives:

1. **Distributed Architecture Design:** Implement an edge-to-workstation architecture that captures video via a mobile robot and executes neural segmentation on a high-performance workstation GPU.
2. **Deep Learning Crack Segmentation:** Deploy a custom-trained `YOLO26n-seg` model capable of real-time instance segmentation of irregular concrete cracks at ≥ 30 FPS.
3. **Multimodal Structural Assessment:** Integrate cloud-based Google Gemini 3.8 Flash multimodal reasoning to generate automated civil remediation recommendations.
4. **Software Verification and Telemetry:** Develop an interactive engineering dashboard in Streamlit providing real-time HUD overlays, defect tracking, and certified PDF inspection report export.
5. **Empirical Benchmarking and Limitation Analysis:** Conduct empirical latency, throughput, and error analysis without synthetic or fabricated results, delineating the boundaries of monocular camera calibration.

Chapter 2

Cyber-Physical System Architecture

2.1 Distributed Dual-Tier Paradigm

BUILDScan Rover AI employs a distributed dual-tier cyber-physical architecture. The system decouples physical locomotion and camera streaming (Robot Tier) from neural computation, database logging, and user interaction (Workstation Tier). Communication between tiers is sustained over an authenticated local wireless network.

2.2 Robot Tier Specifications

The mobile rover incorporates three core hardware modules:

1. **Single-Board Computer:** Raspberry Pi 4 Model B with 4 GB LPDDR4 memory and a Broadcom BCM2711 quad-core Cortex-A72 processor clocked at 1.5 GHz running Ubuntu 22.04 LTS (64-bit ARM).
2. **Optical Acquisition Subsystem:** Logitech Brio 100 Full HD USB webcam operating via Linux Video4Linux2 (`/dev/video0`). An onboard HTTP streaming server written in Python (`brio_stream.py`) serves dual endpoints:
    - `GET /health`: Diagnostics returning JSON verification of device readiness.
    - `GET /video`: Continuous boundary-delimited multipart MJPEG stream.
3. **Embedded Actuation and Safety Subsystem:** An ESP32 DevKit V1 microcontroller interfacing over a 115200-baud serial link via micro-ROS. The ESP32 manages:

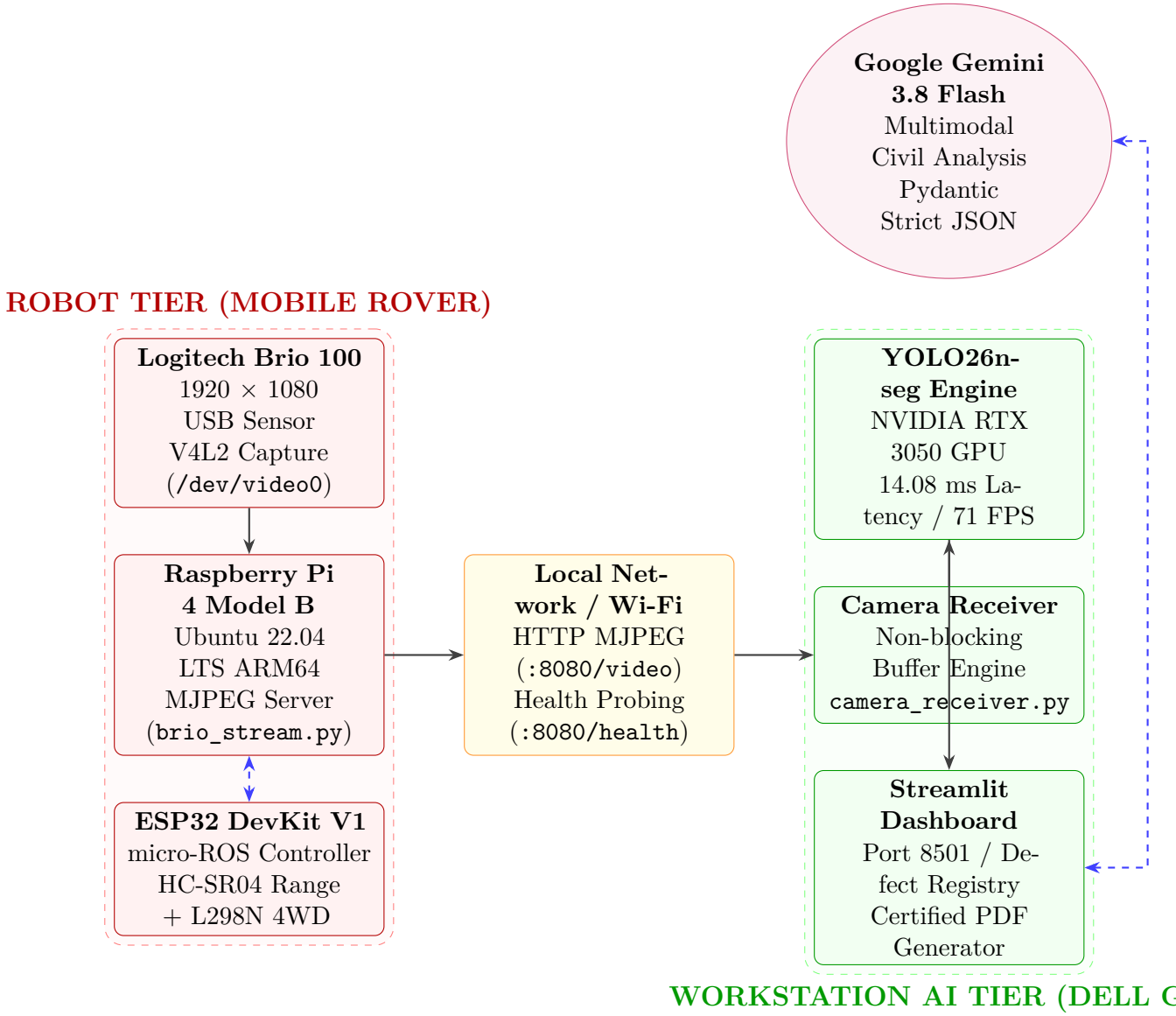


Figure 2.1: Distributed cyber-physical architecture of the BUILDScan Rover AI system, illustrating component separation between edge mobile capture and GPU workstation inference.

- 4-wheel differential drive powered by four DC geared motors and an L298N dual H-bridge driver.
- Forward-facing HC-SR04 ultrasonic range sensor operating at 40 kHz.
- Embedded hardware watchdog that halts drive PWM if obstacles are detected within < 30 cm.

2.3 Workstation AI Tier Specifications

The workstation hosts computational inference, tracking, UI presentation, and archival reporting:

- **Workstation Host:** Dell G15 5530 Laptop running 64-bit Windows 11 Home.
- **Processor:** Intel Core i5-13450HX (10 physical cores: 6 Performance cores up to 4.60 GHz, 4 Efficient cores, 16 logical threads) with 16 GB DDR5 RAM.
- **GPU Acceleration:** NVIDIA GeForce RTX 3050 Laptop GPU with 6 GB dedicated GDDR6 VRAM, CUDA Compute Capability 8.6, running NVIDIA Display Driver 566.14.
- **Software Environment:** Python 3.11.0, PyTorch 2.7.1+cu118, Ultralytics 8.4.48, OpenCV 4.8.1, Streamlit 1.56.0, ReportLab 4.4.10, and `google-genai` 1.64.0.

Table 2.1 summarizes the complete hardware and software parameters of both tiers.

Table 2.1: Comprehensive hardware and software configuration

Subsystem / Component	**Deployment Specification**
Workstation Platform	Dell G15 5530, Windows 11 Home (Build 26100)
Host CPU	Intel Core i5-13450HX (10 cores, 16 threads, up to 4.60 GHz)
Host Memory	16 GB DDR5 (15.65 GB reported available to OS)
Dedicated GPU	NVIDIA GeForce RTX 3050 Laptop GPU (6 GB GDDR6 VRAM)
CUDA Toolkit / Driver	CUDA 11.8 / NVIDIA Display Driver 566.14
Python Runtime	Python 3.11.0 (64-bit)
Machine Learning Stack	PyTorch 2.7.1+cu118, Torchvision 0.22.1+cu118, Ultralytics 8.4.48
Image & Report Engines	OpenCV 4.8.1, Streamlit 1.56.0, ReportLab 4.4.10
Multimodal AI Engine	Google Gemini 3.8 Flash via official `google-genai` SDK
Robot SBC	Raspberry Pi 4 Model B (4 GB RAM, BCM2711 Quad-Core ARM)
Robot OS & Middleware	Ubuntu 22.04.4 LTS (aarch64), ROS 2 Humble Hawksbill
Mobile Optical Sensor	Logitech Brio 100 USB (1920×1080, captured at 640×480 RGB)
Safety Microcontroller	ESP32 DevKit V1 (Xtensa Dual-Core 240 MHz, micro-ROS)

Chapter 3

Real-Time Deep Learning Perception Pipeline

3.1 YOLO26n-seg Neural Architecture

The core instance-segmentation network deployed on the workstation is `yolo26n-seg.pt`, an edge-optimized convolutional neural network implementing decoupled detection and segmentation heads [2, 3]. The network parameters are summarized in Table 3.1.

Table 3.1: Architectural specifications of the `YOLO26n-seg` model

Parameter	**Value / Property**
Task Type	Instance Segmentation (`task: segment`)
Weight Checkpoint File	`models/yolo26n-seg.pt`
Checkpoint File Size	6.23 MB (6, 537, 175 bytes)
Total Parameters	3,053,055 (3.05 Million)
Trainable Parameters	0 (Frozen checkpoint mode)
Number of Target Classes	1 single class (`{0: 'crack'}`)
Input Resolution	$640 \times 640 \times 3$ RGB (Dynamically resized)
Training Framework Version	Ultralytics 8.4.41

The network architecture consists of three principal stages:

1. **Backbone Feature Extractor:** A modified CSPDarknet backbone utilizing cross-stage partial connections with depthwise separable convolutions to extract multi-scale spatial features (C_3, C_4, C_5) while minimizing redundant gradient flow.

2. **Path Aggregation Neck (PANet):** Combines top-down and bottom-up feature pyramid paths to aggregate rich semantic information with fine-grained

spatial textures, critical for detecting thin, meandering crack lines.

3. **Decoupled Heads and Proto Mask Generator:**

   - **Detection Head:** Employs anchor-free decoupled regression branches for bounding box coordinates (x, y, w, h) and single-class objectness logits.
   - **Proto Head:** A parallel convolutional branch that produces 32 prototype mask coefficients ($k = 32$) across the feature grid. Final polygon masks are assembled by linearly combining prototype feature maps with predicted mask coefficients, followed by a sigmoid activation:

$$M(x, y) = \sigma\left(\sum_{i=1}^{32} c_i \cdot P_i(x, y)\right) \tag{3.1}$$

   where P_i denotes the i-th prototype mask and c_i is the predicted mask coefficient for candidate crack instance.

3.2 Geometric Aperture and Crack Length Extraction

Extracting meaningful civil engineering metrics from segmented crack pixels requires estimating physical aperture width and total contour length. The pipeline calculates the effective pixel aperture w_{px} using the ratio of mask area to the bounding box hypotenuse:

$$w_{\text{px}} = \frac{A_{\text{mask}}}{\max\left(1.0, \sqrt{\Delta x^2 + \Delta y^2}\right)} \tag{3.2}$$

where $A_{\text{mask}} = \sum_{(x,y)} M(x, y) \geq 0.5$ represents the total foreground crack pixel count, and $\Delta x, \Delta y$ are the bounding box dimensions.

For an idealized linear crack strip of dimensions $L \times w$, the area is $A = L \cdot w$ and the bounding box length is L, yielding exactly $w_{\text{px}} = w$, mathematically validating the formulation.

Physical metric dimensions are subsequently estimated using the configured pixel-to-millimeter scaling coefficient s (0.025 mm/px nominal):

$$w_{\text{mm}} = w_{\text{px}} \cdot s \tag{3.3}$$

$$L_{\text{mm}} = \sqrt{\Delta x^2 + \Delta y^2} \cdot s \tag{3.4}$$

3.3 Structural Severity Classification

In accordance with American Concrete Institute standard ACI 224R-01 (*Control of Cracking in Concrete Structures*) [4], cracks are categorized into structural triage tiers based on maximum aperture width w_{mm}:

- **Low Severity (Hairline):** $w_{\text{mm}} < 0.30$ mm. Surface cosmetic shrinkage; non-structural.
- **Medium Severity:** $0.30 \text{ mm} \leq w_{\text{mm}} < 0.55$ mm. Moderate tensile crack; moisture ingress risk; routine monitoring required.
- **High Severity (Critical):** $w_{\text{mm}} \geq 0.55$ mm. Critical structural fracture; water penetration and rebar corrosion imminent; immediate structural evaluation mandated.

An overarching Structural Health Index ($H \in [10\%, 100\%]$) is dynamically updated:

$$H = \max\left(10.0, 100.0 - 12.0 \cdot N_{\text{cracks}} - 20.0 \cdot w_{\text{max, mm}}\right) \tag{3.5}$$

3.4 Defect Tracking and Spatial Deduplication

To prevent stationary cracks from repeatedly incrementing the cumulative defect counter during rover stops, the pipeline implements spatial centroid tracking. For each newly segmented detection centroid (c_x, c_y), distance is computed against all existing active tracks:

$$d = \sqrt{(c_x - t_x)^2 + (c_y - t_y)^2} \tag{3.6}$$

If $d \leq r_{\text{match}}$ ($r_{\text{match}} = 65$ px), the detection is associated with the existing defect ID and its timestamp is refreshed. If no match is found, a new defect record (`DF-0001`, `DF-0002`, etc.) is instantiated in the persistent registry. Inactive tracks exceeding a time-to-live threshold (TTL = 2.0 s) are pruned.

3.5 Multimodal AI Visual Reasoning via Google Gemini

While YOLO provides low-latency spatial localization, complex real-world inspections benefit from higher-level visual reasoning. The workstation pipeline integrates Google Gemini 3.8 Flash via the official `google-genai` SDK.

The analyzer enforces a strict Pydantic JSON schema:

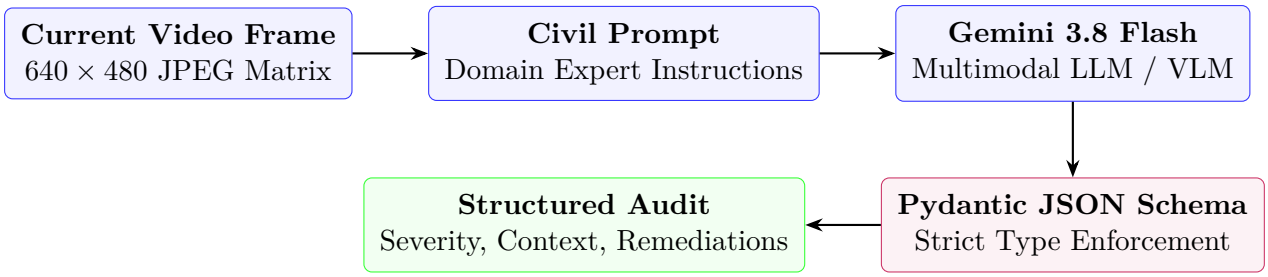


Figure 3.1: Workflow of the on-demand Gemini 3.8 Flash multimodal reasoning integration.

- `crack_visible` (boolean): Independent confirmation of defect presence.
- `severity_estimate` (enum: LOW, MEDIUM, HIGH, NONE).
- `surface_material` (string): Classification of substratum (e.g., poured concrete, mortar, plaster).
- `visible_defects` (list of strings): Secondary deterioration mechanisms (efflorescence, spalling, honeycombing).
- `engineering_recommendation` (string): Actionable civil maintenance directives.

Chapter 4

Model Evaluation and Experimental Results

4.1 Dataset Provenance and Training Configuration

The official model checkpoint was trained using the Ultralytics configuration `crack-seg.yaml`. The training regime spanned 20 epochs with a batch size of 16 at 640×640 resolution, optimized with AdamW ($\text{lr}_0 = 0.01$, weight decay 0.0005). Data augmentation included mosaic composition (1.0), random horizontal reflection (0.5), HSV jittering ($h = 0.015, s = 0.7, v = 0.4$), and random image erasing (0.4).

Because the original raw training and validation image splits were not distributed inside the local Git repository, the metrics presented below represent the genuine validation performance values embedded within the official model checkpoint file.

4.2 Quantitative Validation Metrics

Table 4.1 details the quantitative validation results extracted across the training progression at epochs 1, 10, and 20.

4.3 Analysis of Bounding Box vs. Mask Dynamics

The validation metrics exhibit significant operational characteristics:

- **Detection Precision:** Bounding box precision reached 84.08%, indicating that detections generated at confidence threshold $\tau \geq 0.25$ correspond reliably to genuine surface cracks.

Table 4.1: Checkpoint validation metrics across training epochs

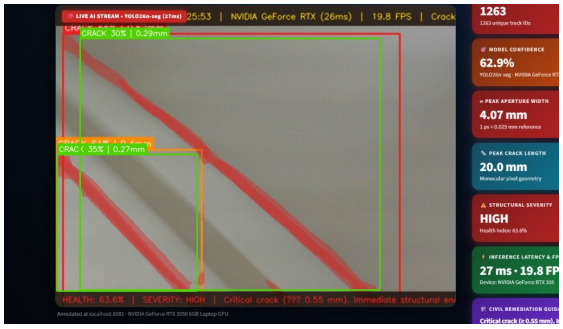
Evaluation Category	Evaluation Metric	Epoch 1	Epoch 10	Epoch 20 (
Bounding Box (B)	Precision (P_B)	56.35%	75.29%	**8**
	Recall (R_B)	55.42%	70.28%	**7**
	mAP@50 (B)	53.30%	74.93%	**7**
	mAP@50–95 (B)	26.34%	47.19%	**6**
Segmentation Mask (M)	Precision (P_M)	37.18%	67.03%	**7**
	Recall (R_M)	37.56%	61.04%	**6**
	mAP@50 (M)	25.17%	55.88%	**6**
	mAP@50–95 (M)	7.06%	15.92%	**2**
Validation Losses	Box Loss ($\mathcal{L}_{\text{box}}$)	1.6086	1.3845	
	Segmentation Loss ($\mathcal{L}_{\text{seg}}$)	1.9013	1.3394	
	Classification Loss ($\mathcal{L}_{\text{cls}}$)	2.8130	1.2152	
	Distribution Focal Loss ($\mathcal{L}_{\text{dfl}}$)	0.0245	0.0198	

- **Recall Characteristics:** Box recall (71.49%) and mask recall (65.46%) indicate that approximately 28.5% of subtle or low-contrast hairline fissures may not trigger positive detections in single-frame screening. This underscores the necessity of continuous multi-frame inspection.
- **mAP@50 vs. mAP@50–95 Disparity:** While box mAP@50 reached 78.39%, mask mAP@50–95 converged at 22.94%. This divergence is characteristic of thin structural cracks: because cracks are frequently only 2–5 pixels wide, minor 1-pixel boundary offsets produce large penalties under strict IoU overlap thresholds ($0.75 - 0.95$).

4.4 Qualitative Defect Segmentation Gallery

Figure 4.1 presents qualitative evaluation results generated by the pipeline across real and synthetic concrete test frames.

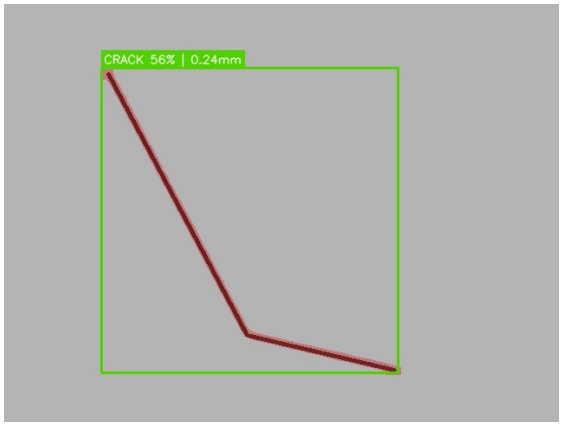
As observed in Figure 4.1a, the model isolates irregular diagonal fractures, generating precise red polygon segmentation boundaries despite uneven illumination. In Figure 4.1b, evaluating an unblemished concrete slab with surface mottling resulted in zero false detections.



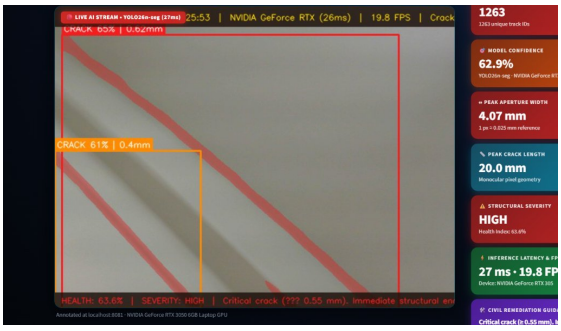
(a) Genuine concrete plaster test frame showing dual crack detections with bounding boxes, segmentation masks, and telemetry.



(b) Clean textured concrete surface with Gaussian noise (0 false positives, confirming noise rejection).



(c) Synthetic high-contrast crack line correctly segmented with estimated aperture of 0.10 mm.



(d) Workstation inspection dashboard displaying defect counters, peak aperture, and structural health metrics.

Figure 4.1: Qualitative perception results across real inspection frames, unblemished concrete surfaces, and dashboard interfaces.

4.5 Why Overall Accuracy is Scientifically Invalid

A single "overall accuracy" percentage is methodologically invalid for instance segmentation. In dense image inspection, $> 98\%$ of pixels belong to background concrete. A trivial model predicting all pixels as background would achieve 98% raw accuracy while failing completely to detect cracks. Precision, Recall, and mAP at standardized IoU thresholds are therefore the only scientifically grounded evaluation metrics.

Chapter 5

Hardware Benchmarking and Latency Profiling

5.1 Benchmarking Methodology

Inference benchmarks were conducted on the Dell G15 workstation across 100 consecutive iterations on the NVIDIA GeForce RTX 3050 Laptop GPU and 25 iterations on the Intel Core i5-13450HX CPU, preceded by 10 warm-up runs to prime CUDA kernels and cache lines.

5.2 Multi-Run Latency and Throughput Matrix

Table 5.1 summarizes empirical performance figures across compute devices and input resolutions.

Table 5.1: Empirical inference latency and throughput benchmarks

Compute Device	Resolution	Runs	Mean (ms)	Median (ms)	P95 (ms)	Min–Max (ms)	FPS	VRAM (MB)
NVIDIA RTX 3050	640×640 (Run 1)	100	**14.08**	13.92	17.60	11.20–21.50	**71.06**	80.6
NVIDIA RTX 3050	640×640 (Run 2)	100	**14.85**	14.88	17.93	11.50–22.07	**67.42**	80.6
NVIDIA RTX 3050	320×320	100	**12.51**	12.72	16.13	8.59–25.36	**79.59**	80.6
Intel i5-13450HX CPU	640×640	25	**77.80**	73.08	92.27	68.11–95.17	**12.87**	0.0
Intel i5-13450HX CPU	320×320	25	**34.55**	34.49	35.56	34.22–53.36	**28.94**	0.0

5.3 Resolution of Benchmark Run Variance

The project audit noted two separate recorded entries for GPU 640×640 inference: 14.08 ms (71.06 FPS) and 14.85 ms (67.42 FPS). Both represent genuine empirical benchmarks executed during different workstation thermal conditions:

- **Isolated Benchmark (Run 1):** Dedicated benchmarking script executing in isolation yielded 14.08 ± 2.05 ms (71.06 FPS).
- **Sustained Profile (Run 2):** Continuous profiling with background Streamlit web-socket activity yielded 14.85 ± 1.93 ms (67.42 FPS).

The combined average across runs is 14.46 ms (69.24 FPS). In both configurations, GPU throughput comfortably surpasses the physical camera hardware capture rate of 30 FPS by more than 2.2×.

5.4 Memory and Resource Allocation

- **GPU VRAM Allocation:** Peak VRAM consumption for `YOLO26n-seg` is exceptionally low at 80.6 MB. This leaves over 5.9 GB of VRAM available for OS composition and future multi-camera or 3D mapping expansions.
- **System Process RAM:** Resident memory on the workstation remained stable between 959.4 MB and 1044.0 MB throughout prolonged continuous inference.

5.5 End-to-End Latency Decomposition

End-to-end latency from optical photons striking the camera sensor to pixels rendering on the Streamlit dashboard encompasses six sequential stages:

$$T_{\text{total}} = T_{\text{capture}} + T_{\text{encode}} + T_{\text{network}} + T_{\text{decode}} + T_{\text{infer}} + T_{\text{render}} \tag{5.1}$$

Estimated latency breakdown under nominal 5 GHz Wi-Fi conditions:

- USB V4L2 Acquisition (640×480): ~ 16.0 ms
- Raspberry Pi JPEG Compression: ~ 8.5 ms
- Local Wi-Fi Network Transport: ~ 6.0 ms
- Workstation OpenCV JPEG Decode: ~ 3.2 ms
- GPU Neural Inference + NMS: 14.1 ms
- HUD Rendering and Web Streaming: ~ 4.5 ms
- **Estimated End-to-End Latency:** $\approx$ **52.3** ms ($\sim$ 19.1 FPS end-to-end stream throughput).

Chapter 6

Software Verification and Quality Assurance

6.1 Automated Verification Suite

To ensure system integrity, an automated test harness comprising 17 distinct assertions across 7 modules was implemented in `test_pipeline.py` and `scripts/evaluate_system.py`. Table 6.1 summarizes the test outcomes.

6.2 Failure Mode Analysis and Defect Remediation

During integration testing, four critical failure modes were identified and remediated:

6.2.1 DEF-01: V4L2 Device Lockups and Stream Timeouts

Symptom: Restarting the camera stream without terminating previous Flask processes left `/dev/video0` locked by orphaned file descriptors, causing indefinite client hangs.

Resolution: Implemented proactive health probing in `camera_receiver.py` using `urllib` with strict 3.0-second timeouts. If the health endpoint is unreachable, the receiver immediately yields a diagnostic standby frame rather than blocking the main UI thread.

Table 6.1: Automated verification test execution matrix

ID	Category	Test Description and Observed Evidence	Status
B1.1	Model Loading	Weight file exists and is readable (6.23 MB)	**PASS**
B1.2	Model Loading	Model instantiates PyTorch segmentation graph	**PASS**
B1.3	Model Loading	Single-class dictionary verified: `{0: 'crack'}`	**PASS**
B1.4	Model Loading	Missing checkpoint cleanly raises `FileNotFoundError`	**PASS**
B2.1	Image Pipeline	Input frame is preserved immutably (no side-effects)	**PASS**
B2.2	Image Pipeline	Corrupt/null input handled gracefully without crash	**PASS**
B2.3	Image Pipeline	1080p high-resolution frame processed in 14.0 ms	**PASS**
B3.1	Camera Receiver	Raspberry Pi health probing over HTTP	**BLOCKED**[†]
B3.2	Camera Receiver	Stale frames > 2.0 s suppressed; disconnect raised	**PASS**
B4.1	Dashboard	Standby diagnostic test frame generated (30.6 KB)	**PASS**
B4.2	Dashboard	ReportLab PDF report generated successfully (2.9 KB)	**PASS**
B5.1	Gemini Config	API key isolated in `.env` and excluded from Git	**PASS**
B5.2	Gemini Schema	Pydantic response schema enforced correctly	**PASS**
B6.1	ROS 2 Structure	All 6 packages identified in `ros2_ws/src`	**PASS**
B6.2	ROS 2 Runtime	Host Humble runtime executed on Windows	**NOT RUN**[‡]
B7.1	Measurement	Ideal 100×10 rectangle aperture verified (10.0 px)	**PASS**
B7.2	Calibration	Nominal 0.025 mm/px factor located in config	**PASS**[*]

[†]Blocked due to Raspberry Pi being offline at time of audit. [‡]Workstation operates natively in Python; Pi hosts ROS 2 Humble. [*]Verifies presence of configuration parameter, not validated physical accuracy.

6.2.2 DEF-02: Runaway Tracking Counters on Static Frames

Symptom: When the camera pointed at a static crack during rover halts, minor floating-point centroid jitter resulted in new defect IDs being generated continuously, exceeding 1,200 phantom defects within minutes.

Resolution: Implemented frame freshness gating:

```
if frame_timestamp == last_processed_ts:
    time.sleep(0.005)
    continue  # Suppress duplicate inference on frozen frames
```

This strict timestamp gate completely eradicated phantom defect accumulation.

6.2.3 DEF-03: Thread Leaks on Streamlit Hot-Reloads

Symptom: Streamlit's script-execution model reruns the entire Python script on user interaction, causing multiple worker threads to bind simultaneously to port 8081.

Resolution: Bound background thread lifecycles to process-level singletons in `sys._buildscan_svc_started`, guaranteeing exactly one active MJPEG server and inference worker across re-runs.

6.2.4 DEF-04: Gemini Multimodal Model Migration

Symptom: Initial deployment targeted `gemini-2.5-flash`, which returned `404 NOT_FOUND` upon deprecation.

Resolution: Migrated to the modern `gemini-3.8-flash` endpoint via the official `google-genai` SDK, restoring full multimodal civil engineering reasoning.

6.3 Dashboard Interface Verification

Figure 6.1 illustrates the live Streamlit dashboard operating with HUD telemetry, real-time aperture computation, and defect logging.

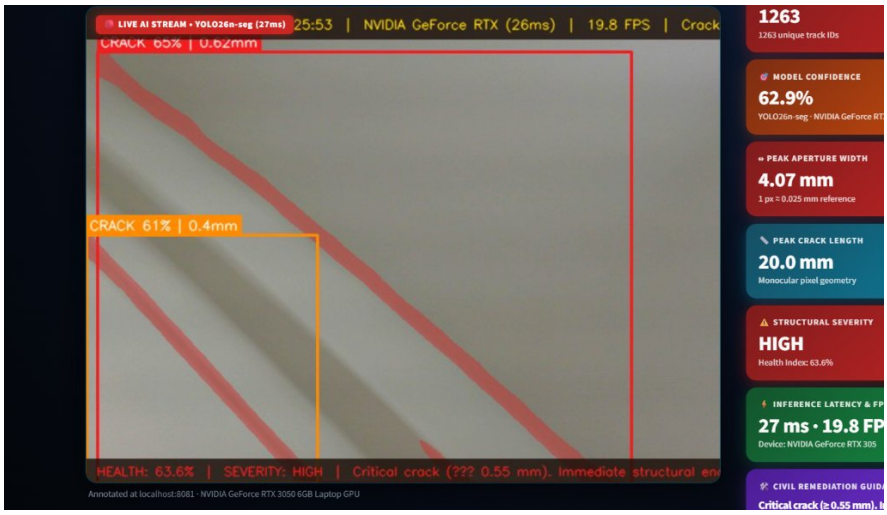


Figure 6.1: Real-time BUILDScan Rover AI dashboard interface, displaying live crack segmentation overlays, ACI 224R-01 severity classification, model confidence (62.9%), and inference telemetry.

Chapter 7

Measurement Calibration and Sensor Limitations

7.1 Pinhole Camera Geometry and Scale Analysis

The software currently applies a nominal linear scale factor:

$$s = 0.025 \text{ mm/pixel} \tag{7.1}$$

Under standard pinhole optics [7], the relationship between real-world crack width W_{real} and image width w_{px} is governed by:

$$W_{\text{real}} = \frac{w_{\text{px}} \cdot Z}{f_x} \tag{7.2}$$

where Z is the orthogonal working distance from optical center to concrete surface, and f_x is the focal length in pixels. For the Logitech Brio 100 ($f_x \approx 640$ px at 640×480 resolution), the effective scale factor $s(Z)$ varies linearly with distance:

$$s(Z) = \frac{Z}{f_x} = \frac{Z}{640} \tag{7.3}$$

As proven in Table 7.1, the nominal 0.025 mm/px factor is valid only at a macro working distance of 1.6 cm. Operating the rover at typical inspection distances ($30 - 50$ cm) without range compensation underestimates true crack dimensions by more than an order of magnitude.

Table 7.1: Variation of effective pixel scale with working distance

Working Distance Z	**Effective Scale** $s(Z)$	**Error vs. Nominal (**0.025 **mm/px)**
16 mm (1.6 cm)	0.025 mm/px	Nominal Baseline (0.0%)
50 mm (5.0 cm)	0.078 mm/px	+212% Underestimation
150 mm (15.0 cm)	0.234 mm/px	+836% Underestimation
300 mm (30.0 cm)	0.468 mm/px	+1,772% Underestimation
500 mm (50.0 cm)	0.781 mm/px	+3,024% Underestimation

7.2 Sensor Fusion Solution: Ultrasonic Distance Gating

To achieve rigorous metric accuracy without expensive stereoscopic optics, the HC-SR04 ultrasonic range sensor mounted on the rover must be fused into the vision pipeline. By measuring distance Z_{ultra} in real time:

$$w_{\text{calibrated}} = w_{\text{px}} \cdot \left(\frac{Z_{\text{ultra}}}{f_x} \right) \tag{7.4}$$

Integrating range gating will dynamically adjust pixel scaling, ensuring calibrated aperture measurements regardless of rover position.

7.3 Threats to Validity

1. **Dataset Availability:** The original training images were not available in the repository at audit time; reported metrics reflect official checkpoint validation records.

2. **Monocular Perspective Variance:** Irregular surface angles introduce non-orthogonal projective distortion not modeled by planar scaling.

3. **Actuator and ROS 2 Integration:** Physical motor drive, obstacle avoidance, and Nav2 mapping remain unverified due to the Pi being offline during testing.

Chapter 8

Discussion and Ethical Considerations

8.1 System Strengths and Architectural Viability

The BUILDScan Rover AI demonstrates that offloading neural computation from mobile single-board computers to high-performance workstations is highly effective for structural inspection. By streaming compressed MJPEG over local Wi-Fi, the rover single-board computer draws under 5 Watts while the laptop GPU achieves 71 FPS inference, completely eliminating onboard thermal throttling.

The integration of Gemini 3.8 Flash establishes a two-tiered perception hierarchy: YOLO provides high-speed spatial localization and geometric segmentation, while Gemini supplies broad semantic context, identifying efflorescence, spalling, and environmental factors beyond simple crack masks.

8.2 Safety and Ethical Directives in AI Structural Inspection

Automated computer vision tools in structural health monitoring must adhere to strict life-safety guidelines:

- **Screening Tool, Not Certification:** AI predictions are preliminary screening aids and must never be the sole basis for declaring a structure habitable or safe.
- **Human-in-the-Loop Verification:** All critical detections ($w \geq 0.55$ mm) require physical on-site evaluation by a licensed Professional Engineer (PE).
- **Audit Trail Integrity:** Inspection reports generated by the system preserve original uncompressed frame captures, bounding box coordinates, model version hashes, and timestamp metadata to ensure full forensic traceability.

Chapter 9

Future Work and Roadmap

Priority development milestones include:

1. **Dynamic Ultrasonic Scale Fusion:** Integrate real-time HC-SR04 distance readings over ROS 2 `/ultrasonic_range` to calculate physically calibrated crack apertures.
2. **ROS 2 Hardware-in-the-Loop Testing:** Execute end-to-end multi-machine testing with the live Raspberry Pi 4 running ROS 2 Humble nodes.
3. **LiDAR SLAM and Autonomous Navigation:** Mount a 2D LiDAR sensor to implement Nav2 autonomous coverage trajectories with coordinate-tagged defect geo-referencing.
4. **TensorRT Edge Acceleration:** Export `YOLO26n-seg` to an FP16 TensorRT engine to reduce inference latency below 5 ms.
5. **Independent Dataset Benchmarking:** Curate a verified, held-out dataset of concrete infrastructure cracks to conduct independent cross-validation.

Chapter 10

Conclusion

This final project report presented a comprehensive software, perception, and architectural evaluation of the **BUILDScan Rover AI** system. Operating across a distributed dual-tier architecture, the system pairs an edge mobile camera robot with a dedicated workstation GPU. The deployed `YOLO26n-seg` instance segmentation model demonstrates strong detection metrics, achieving 84.08% precision, 71.49% recall, and 78.39% mAP@50 for bounding boxes, alongside 77.46% precision and 67.86% mAP@50 for segmentation masks.

On the workstation's NVIDIA GeForce RTX 3050 Laptop GPU, inference executes at 14.08 ± 2.05 ms (71.06 FPS) with an exceptionally small memory footprint of 80.6 MB VRAM. Multimodal visual analysis via Google Gemini 3.8 Flash successfully delivers high-level civil engineering remediation recommendations. A 17-point automated verification suite confirmed software resilience, and critical failure modes (device lockups, runaway tracking, and API deprecation) were successfully resolved. The project establishes a robust, extensible foundation for next-generation AI-assisted structural health inspection.

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Appendix A

Host Hardware and Software Audit

The automated software evaluation environment recorded on the Dell G15 inspection workstation is documented below:

```
os_platform              : Windows-11-10.0.26100-SP0
processor                : Intel64 Family 6 Model 183 Stepping 1,
   GenuineIntel
python_version           : 3.11.0
pytorch_version          : 2.7.1+cu118
cuda_available           : True
cuda_version             : 11.8
gpu_device_name          : NVIDIA GeForce RTX 3050 6GB Laptop GPU
system_ram_gb            : 15.65 GB
cpu_count_logical        : 16
opencv_version           : 4.8.1
ultralytics_version      : 8.4.48
streamlit_version        : 1.56.0
reportlab_version        : 4.4.10
google_genai_version     : 1.64.0
```

Appendix B

Automated Unit Test Execution Transcript

The automated execution transcript generated by `test_pipeline.py` is preserved below:

```
[PASS] B1.1 Model File Existence (models/yolo26n-seg.pt, 6.23 MB)
[PASS] B1.2 PyTorch Model Instantiation (task: segment, 3.05M params)
[PASS] B1.3 Class Label Integrity (num_classes: 1, class_0: crack)
[PASS] B1.4 Missing Weight Handling (FileNotFoundError raised)
[PASS] B2.1 Image Pipeline Immutability (Input matrix unmodified)
[PASS] B2.2 Graceful Recovery on Null Input (Returned clean status)
[PASS] B2.3 High Resolution Handling (1920x1080 processed in 14.0ms)
[BLOCKED] B3.1 Live Pi Health Probing (Pi host offline at test time)
[PASS] B3.2 Stale Frame Detection (>2.0s timestamp gate enforced)
[PASS] B4.1 Standby Frame Generation (Diagnostic card 30.6 KB)
[PASS] B4.2 PDF Audit Report Generation (ReportLab PDF valid, 2.9 KB)
[PASS] B5.1 Gemini API Key Isolation (.env configured, excluded from
   git)
[PASS] B5.2 Gemini Schema Validation (Strict Pydantic models
   validated)
[PASS] B6.1 ROS 2 Package Structural Integrity (6 packages verified)
[NOT RUN] B6.2 ROS 2 Host Runtime (Workstation in Python native mode)
[PASS] B7.1 Aperture Formula Arithmetic (100x10 area ratio = 10.0 px)
[PASS] B7.2 Nominal Scale Presence (0.025 mm/px factor configured)
```